

00:12:34

- Main cause of symptomatic hypercalcemia - **Malignancy** (Squamous cell carcinoma of the lung)
- 1st cause of asymptomatic hypercalcemia - **Parathyroid adenoma**
- First-line management of dangerous hypercalcemia → **Normal saline** (Normal calcium: 9-11 mg%)
- First-line management of dangerous Hypermagnesemia → **Calcium gluconate**
- First-line management of Tetany - **Calcium gluconate**

PRIMARY HYPERPARATHYROIDISM

00:15:38

- Hack to remember features - **STONES, BONES, GROANS AND MOANS**
- Leading cause - **Parathyroid adenoma** (Inferior parathyroid)
- Associated with **MEN1, MEN2/2A**
- ↑ calcium levels (Raised to 15mg %)

CLINICAL FEATURES

00:17:22

- Abdominal pain - Nephrocalcinosis
- Constipation - Due to spasm of sphincters
- Bone pain
- Muscle weakness (proximal)
- Psychic moans - Depression/Psychosis
- Hypercalcemia can cause **systolic arrest at serum calcium level > 15 mg%**

WORK UP

00:20:36

- Serum calcium ↑
- Serum phosphate ↓
- Serum alkaline phosphatase ↑
- Serum PTH assay: Tremendous ↑ in PTH level
- The imaging modality of choice - **Tc99 Sestamibi SPECT/CT**
- X-ray hand: **Subperiosteal resorption of phalanges**

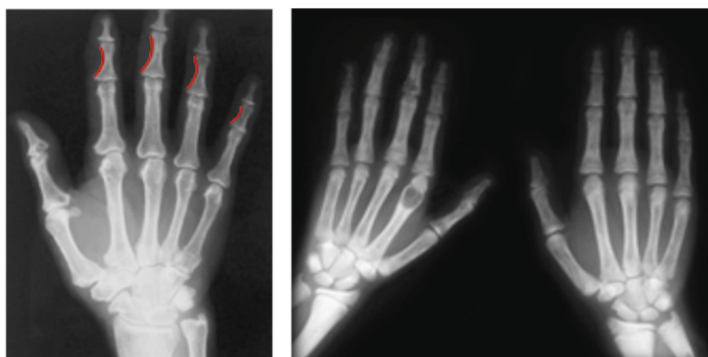
Important Information

- **Tc₉₉ pertechnetate scan** is used in Meckel's diverticulum

TREATMENT

00:22:40

- **Resection of the tumor**



Concavity due to subperiosteal resorption of bone

MUST KNOW CAUSES OF HYPERCALCEMIA

00:23:59

Mnemonic: HAS MILK - INTOXICATION

- **H**: Humoral Hypercalcemia of Malignancy → Squamous cell carcinoma lung/Ca breast
- **A**: Adenoma (Parathyroid)
- **S**: Sarcoidosis (Granulomas: 1 α hydroxylase, synthesis of Vitamin D3) → Steroids
- **M**: Milk alkali syndrome: Antacids abuse, Thiazides, Lithium
- **Intoxication**: Vitamin D3 intoxication → Steroids

OTHER MISCELLANEOUS CAUSES OF HYPERCALCEMIA

00:29:30

- Jansen disease - PTHR1: Gain of function
- Familial Hypocalciuric hypercalcemia - CaSR is present in Parathyroid & Kidney
- CaSR malfunction leads to
 - PTH release $\uparrow$
 - $\uparrow$ Reabsorption of calcium
- **Causes of steroid-responsive hypercalcemia**
 - Vitamin D3 intoxication
 - Sarcoidosis
 - Multiple myeloma
 - Leukemia
 - Lymphoma

MANAGEMENT OF HYPERCALCEMIC CRISIS

00:33:38

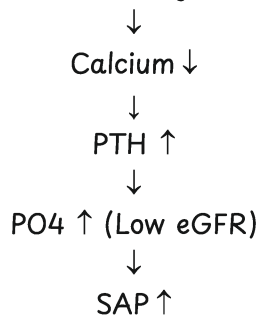
- Normal saline - 1st line
- NS + Furosemide drip (Ca₂₊ loss in urine)
- Zoledronate IV (onset: 1-2 days, effect-3 weeks) - DOC
- Calcitonin nasal spray - Tachyphylaxis, rapid onset
- Denosumab S/C bind to RANK ligands - Inhibit osteoclastic activity
- Oral Phosphate
- Steroids

SECONDARY HYPERPARATHYROIDISM

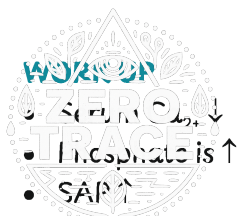
00:38:15

- M/c causes: Chronic kidney disease (Diabetic Nephropathy)
- The kidney is the site of vitamin D₃ production

In patients with CKD, the synthesis of vitamin D₃



- They present with symptoms of bone pain and pathological fractures



00:41:35

- Inorganic phosphate is $\uparrow$
- SAP $\uparrow$
- PTH values $\uparrow\uparrow$
- Vitamin D_3 - $\downarrow$
- eGFR - 30ml/min
- S.Creatinine - $\uparrow$

MANAGEMENT

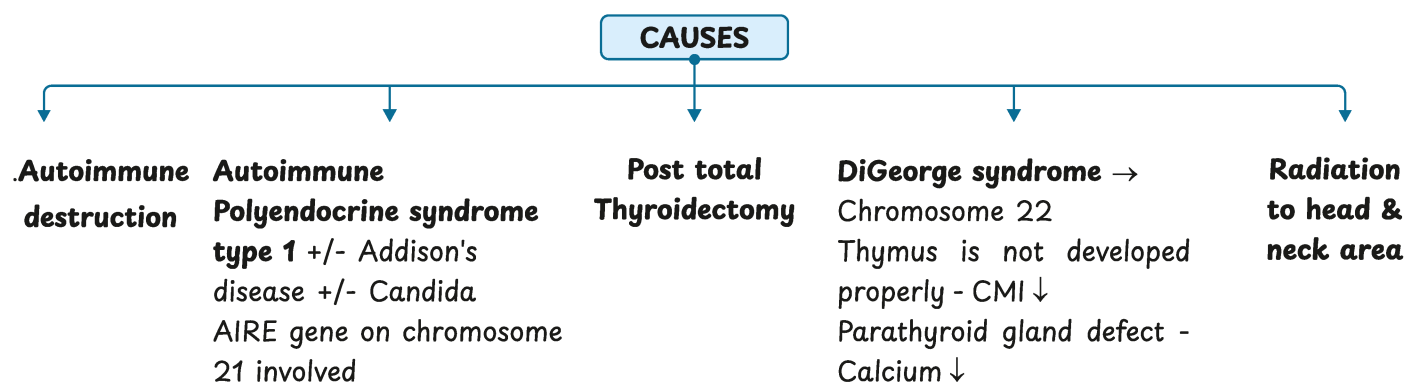
00:42:50

- Kidney transplantation: Definitive treatment
- Calcium, Vitamin D_3
- Role of Cinacalcet
 - Inhibit PTH release
 - Calcimimetic drug - Acts on CaSR of parathyroid gland

HYPOPARATHYROIDISM

00:45:17

CAUSES



- Tetany - Death is due to **laryngospasm**

TETANY CASE SCENARIO

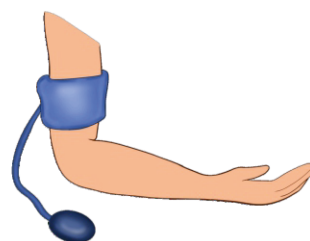
00:50:39

- You are an intern posted in the thyroid surgery unit. A 35-year-old patient who underwent total thyroidectomy three days ago is complaining of perioral and periungual paraesthesias. On examination, the following signs are noted

TROUSSEAU SIGN

00:52:47

- SBP + 20 mmHg - Inflate the cuff → Keep for 3 minutes
→ **Carpopedal spasm/Obstetrician hand**
- Irritability of the median nerve can be demonstrated



CHVOSTEK SIGN

- Tap on the anterior border of the parotid gland
- To demonstrate the facial nerve irritability
- **Manifestation - Circum-oral twitching**



Important Information

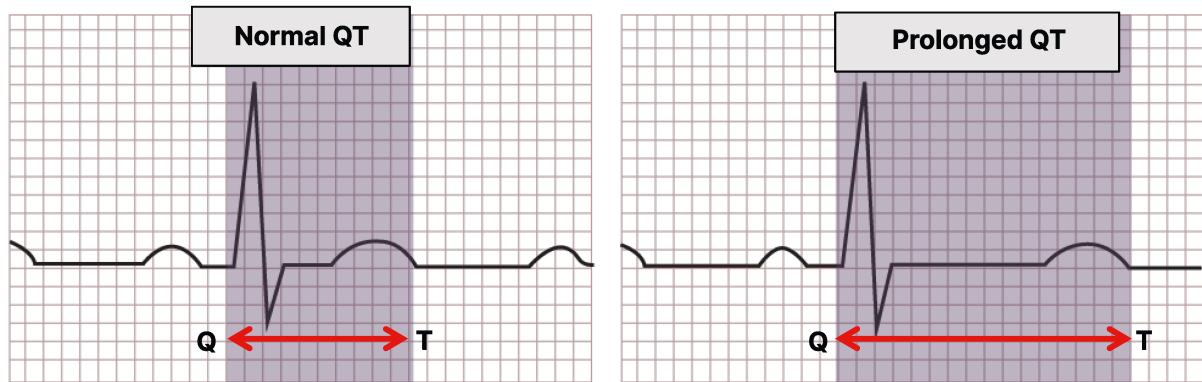
TSS/CSFAS SYNDROME

- Migratory superficial thrombophlebitis → lung, pancreatic, stomach cancer

WORK UP

00:56:26

- ECG: **Prolonged QT interval** (Normal is 360-440 milliseconds)
 - The calcium in the blood is inversely related to the QT interval



- Ionized calcium ↓
- Serum calcium ↓
- Serum albumin - Normal

TREATMENT

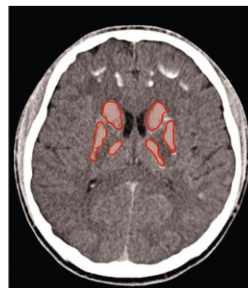
00:58:02

- Emergency management of tetany - **I.V 10% calcium gluconate over 10 minutes**
- Surgically induced hypoparathyroidism - Teriparatide S/c

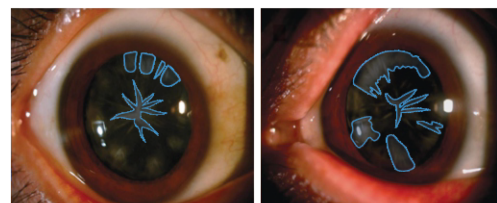
CLINICAL FEATURES

00:59:07

- Causes Basal ganglia calcification
 - Dystonia
 - Oculogyric crisis
 - Hemiballismus
 - Ocular cataracts
 - Abnormal dentition



B/L Basal ganglia calcification



Ocular cataracts

WORK UP

01:01:02

- Calcium ↓
- Phosphate ↑
- SAP : Normal
- Serum PTH assay ↓

TREATMENT

01:02:13

- Teriparatide is given in surgery-induced hypoparathyroidism
- Calcium and vitamin D supplements are given

PSEUDOHYPOPARATHYROIDISM

01:03:22

CAUSE

- A defect in the **GNAS gene** on chromosome 20 leads to an inactivating mutation of GPCR that leads to PTH resistance (PTH assay $\uparrow$)

PATTERN OF INHERITANCE

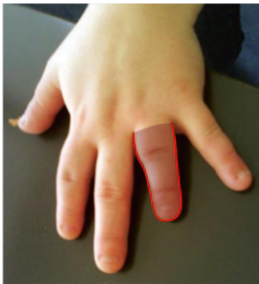
01:04:30

- Genomic imprinting: Inheritance from the maternal side

CLINICAL FEATURES

01:05:11

- **Albright hereditary osteodystrophy (AHO)**
 - Short stature
 - Obesity
 - Brachydactyly (short metacarpal and metatarsal bone)
 - Low I.Q
 - Short 4th and 5th metacarpal bones $\rightarrow$ Knuckle-Knuckle-dimple-dimple sign also known as **Archibald's sign**



Short 4th metacarpal



Short 4th metatarsal



Archibald's sign

01:08:24

WORK UP

- Serum calcium $\downarrow$
- Phosphate $\uparrow$
- SAP is normal
- PTH assay - $\uparrow$
- Urinary cAMP levels are low

TREATMENT

01:10:22

- No definitive treatment

PSEUDO PSEUDO HYPOPARATHYROIDISM (PPHP)

01:10:24

- Inherited from the paternal side
- The action of PTH on bones is defective and has AHO
- Action on kidney is normal - Calcium, phosphate & SAP levels are normal

DISEASE	PTH	CAUSES	PHOSPHATE	CLINICAL FEATURES
Hypoparathyroidism	$\downarrow$	Autoimmune	$\uparrow$	Neurological manifestations
Pseudohypoparathyroidism	$\uparrow$	GNAS gene; chromosome 20, PTH resistance	$\uparrow$	AHO
Pseudo-pseudo hypoparathyroidism	Normal	Paternal inheritance	Normal	AHO



	PRIMARY HYPERPARATHYROIDISM (Cause - Adenoma)	SECONDARY HYPERPARATHYROIDISM SM (Cause-CKD)	HYPOPARATHYROIDISM (Cause -Autoimmune)	Pse PHP
Calcium	↑	↓	↓	↓
Phosphate	↓	↑	↑	↑
SAP	↑	↑	Normal PTH ↓	Normal PTH ↑

NUTRITIONAL RICKETS

01:18:23

CALCIUM	↓ - PTH ↑
PHOSPHATE	↓
SAP	↑
VITAMIN D3	↓
TREATMENT	6 Lac IU Vitamin D3

OSTEOPOROSIS

01:20:04

- Post-menopausal woman with **severe low backache and bone pain**
- Calcium - Normal
- Phosphate - Normal
- SAP - Normal
- **IOC- Dexa scan**
 - T score ≤ -2.5 SD
 - Z score ≤ -2.5 SD

TREATMENT

- Bisphosphonates - Zolendronate/Risedronate (risk of GERD)
 - Usage > 3-5 years increased risk of Atypical/Chalk stick fracture of femur, Osteonecrosis of jaw
- Calcitonin nasal spray
- Teriparatride-Osteosarcoma risk
- Denosumab : Osteoclasts ↑ , Osteoblasts - Normal



VITAMIN D DEPENDENT RICKETS

01:23:41

TYPE 1	TYPE 2
• The gene encodes for 1 hydroxylase is defective	• End-organ resistance to vitamin D₃
• Synthesis of active form of vitamin D is decreased	
• T/t - Vitamin D₃	• T/t - IV calcium infusion



Q. The lady presents with complaints of recurrent episodes of renal colic, constipation, and rib pain. The workup shows low serum calcium with high phosphate and high PTH values. Which explains her current presentation?

- a. PTH and Parathyroid adenoma
- b. PTH-rp and humoral hypercalcemia of malignancy
- c. CKD and metabolic bone disease
- d. Catecholamines and Pheochromocytoma

Ans (c)

Q. A 65-year-old post-menopausal woman presents with a history of multiple fragility fractures over the past few years. Dual-energy X-ray absorptiometry (DEXA scan) reveals low bone mineral density. Laboratory investigations show normal calcium, normal phosphate levels, and normal serum 25-hydroxyvitamin D levels. Which of the following drugs is not effective in the management of this condition?

- a. Teriparatide
- b. Cinacalcet
- c. Denosumab
- d. Bisphosphonates

Ans (b)



MULTIPLE ENDOCRINE NEOPLASIA

MULTIPLE ENDOCRINE NEOPLASIA

- Parathyroid adenoma seen in MEN 1, 2, 4 syndromes

DIAGNOSTIC CRITERIA FOR MEN SYNDROME

00:01:01

- (+) 2/3 endocrine tumors
 - Documentation of familial basis
 - Autosomal dominant/sporadic
 - Genetic analysis
 - Germ line mutation analysis
- } At least one should be (+)

MEN-1

- Chromosomal defect - 11q13
- Gene involved - MEN 1 gene
 - Has role in cell cycle/ growth
- Parathyroid adenoma (M/c)
 - Tumor present in all four parathyroid glands
 - ↑ PTH
 - ↑ Calcium → nephrolithiasis, constipation
 - Damage to kidneys → polyuria, polydipsia
 - T/t - Total parathyroid resection
- Pancreatic adenoma - ZES → PUD
 - PUD - M/c cause of death
 - UGIE - ulcers at 1st, 2nd part duodenum and jejunum
- Pituitary adenoma - Prolactinoma
 - Galactorrhea (male and female)
 - Female: Anovulatory cycles
 - Male: hypogonadism
- Associated conditions - (Mnemonic: A2CP)
 - Angiofibroma
 - Adrenocortical tumors
 - Carcinoid tumor: overproduction of 5HT derivatives (serotonin/ histamine) → flushing episodes
 - Pheochromocytoma/chromaffinoma (< 1%)

MEN-2/ 2A/ SIPPLE SYNDROME

00:10:23

- Chromosome defect: Chromosome 10
- Gene involved: Ret gene/Ret 634



MEN-2 SYNDROME

Parathyroid adenoma

Pheochromocytoma -
leading cause of death

- Medullary Carcinoma of thyroid (M/c)
 - A-cal amyloid deposition
 - Tumor marker - Calcitonin

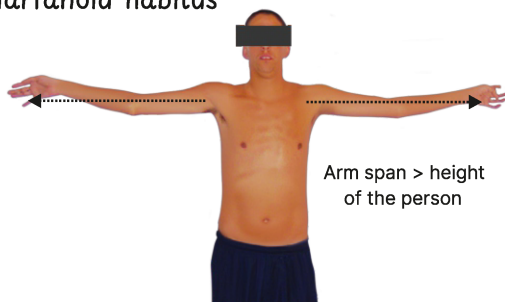
- Associated with Hirschsprung disease (congenital agangliogenesis)
- Features of medullary carcinoma thyroid seen in Sipple syndrome -
 - Presents as neck mass with dysphagia and secretory diarrhea (d/t calcitonin and serotonin)
 - Serum calcium normal in MCT
 - Thyroid scan - cold nodules
 - Isotopes: I-132, I-123
 - Cold nodules suggestive of malignancy
 - Prophylactic total thyroidectomy with levothyroxine supplementation

MEN-3/2B

00:15:31

- Chromosome defect - Chromosome 10
- Gene involved - Ret 918
- Medullary Ca thyroid (M/c)
- Pheochromocytoma - leading cause of death
- M₄
 - Marfanoid habitus
 - Mucosal neuromas (bumpy lips and tongue)
 - Medullary corneal nerve fibers (+)
 - Megacolon

Marfanoid habitus



Mucosal neuromas

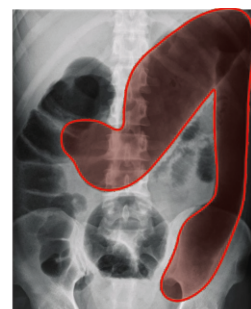


Medullated corneal nerve fibers



Corneal nerve fibers

Megacolon



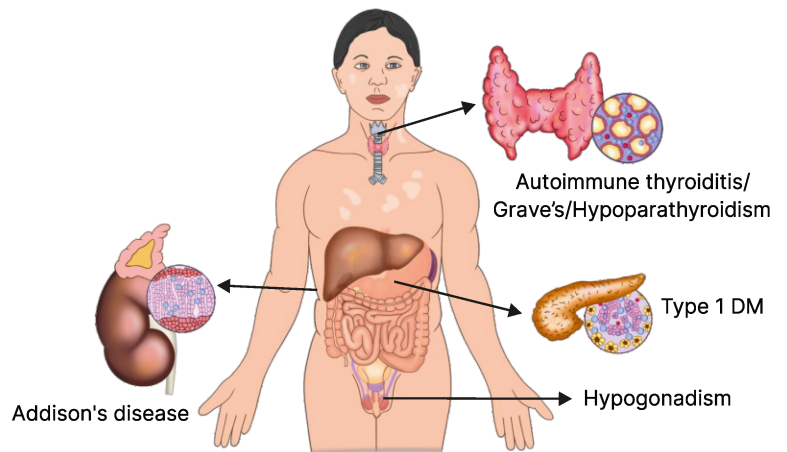
- Chromosome defect - Chromosome 12
- Gene involved - **CDKN1B** gene
- Pituitary adenoma
- Parathyroid adenoma
- Reproductive tract tumor
 - **Male** - Testicular cancer
 - **Female** - Neuroendocrine cervical cancer

AUTOIMMUNE POLYENDOCRINE SYNDROME

00:23:11

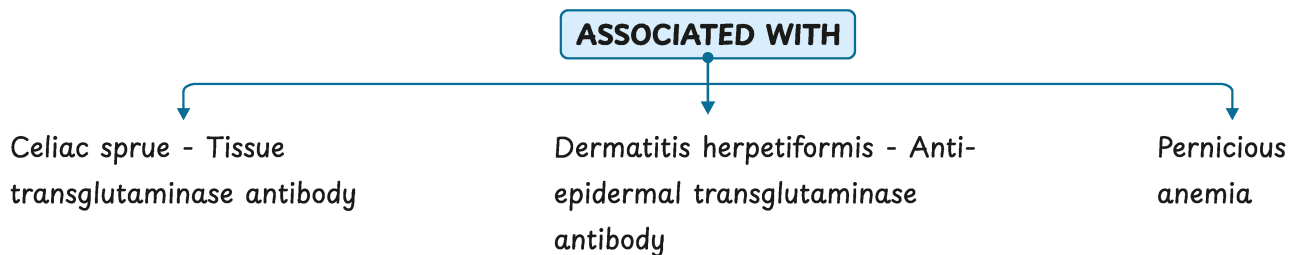
TYPE-1 (Mnemonic - CHAI)

- Mucocutaneous **c**andidiasis
 - Asplenia + ↓ thymus development → ↓ immunity → Candidiasis
- **H**ypoparathyroidism
- **H**ypogonadism
- **H**ypothyroidism: Grave's disease/ Autoimmune thyroiditis
- **A**IRE gene defect on Chromosome 21
- **A**utosomal recessive
- **A**ddison's disease
- Type **1** DM



TYPE-2

- Hypoparathyroidism **not** seen





Q. A 38 year old man with a family history of MEN-2 presents with episodes of severe headache, palpitations, and excessive sweating. His blood pressure is 180/100 mmHg. Which investigations would not be done in this case ?

- a. Serum calcium
- b. Serum calcitonin
- c. Plasma fractionated metanephrine level
- d. Serum prolactin

Ans. (d)

Q. Which of the following is not seen in MEN 3/2B syndrome?

- a. Medullary carcinoma thyroid
- b. Parathyroid adenoma
- c. Mucosal neuroma
- d. Marfanoid habitus

Ans. (b)



13.05.2020 DISEASES OF ANTERIOR PITUITARY

ACROMEGALY

00:00:52

- WWE wrestlers suffering from **Acromegaly**
- Acromegaly is seen after puberty.



ETIOLOGY

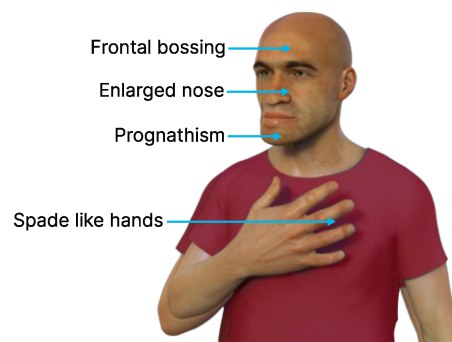
00:01:36

- Somatotroph adenoma → causing overproduction of GH ↑ (**M/c cause**)
- Mixed Mammo somatotroph adenoma – Overproduction of GH & Prolactin (Galactorrhea, Hypogonadism)
- Hypothalamic hamartoma, Choriostoma - loss of inhibitory control over release of GH.
- Extra pituitary source of GHRH → Carcinoid tumor
- Extra pituitary source of GH
- Ectopic pituitary tissue in Nasopharynx
- Pancreatic tumor

CLINICAL FEATURES

00:05:46

- Manifestation seen after puberty
- **Spade like hand**
- Acral enlargement
 - ↑ Gloves size
 - ↑ Finger thickness
 - Shoe size ↑
- Frontal bossing
 - Coarse faces
 - Hyperhidrosis (↑ in sweating)
 - Increased sebum excretion
 - Widened gap between the lower incisors d/t Prognathism
- **Height remains the same**
- ↑ Heel pad thickness: Indirect parameter to assess soft tissue growth
- Weight ↑
 - Arthropathy, Kyphosis, Carpal tunnel syndrome
- **Cardiomyopathy** → CHF, Arrhythmia (**M/c cause of mortality**)
 - Overall life expectancy ↓ by 10 years
 - Higher incidence of stroke & obstructive sleep apnea → Systemic & pulmonary artery HTN



- ↑ incidence of Colonic polyps and malignancy
- Hypertension
- Fasting plasma glucose ↑: 100-125 mg/dL
- 2-hour plasma glucose ↑: 140-199 mg/dL
- Galactorrhea
- ± Visual field defects: Bitemporal hemianopia

Enlarged tongue



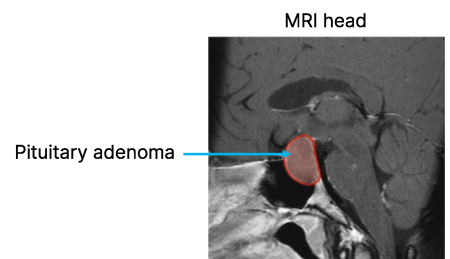
Important Information

- Causes of Galactorrhea
 - Prolactinoma
 - Antipsychotic drugs
 - Chronic renal failure
 - Primary Hypothyroidism
 - Acromegaly

WORK UP

- **Screening test:** Insulin-like growth factor 1 (IGF-1) levels.
- **IOC** - Oral glucose tolerance test
 - 75gm of glucose suppress GH
 - But here there is **failure to suppress GH levels** to < 0.4µg/L
- Prolactin levels: ↑ in mixed adenomas
- MRI head: Localize pituitary adenoma
 - Macroadenoma of size > 10 mm → Visual field deficit
- X-ray foot: Heel pad thickness > **25 mm**

00:18:43



X-ray foot



Increased heel pad thickness

MANAGEMENT

- Preoperative shrinking of tumor → Improve overall performance & quality of life.
 - Lanreotide - Act on SST 2 receptor
 - Pasireotide - Act on SST 5 receptor
- Surgery - **Transsphenoidal surgery**
- **PEGVISOMANT:** Act against GH receptors
 - Indication
 - Recurrence of tumor
 - Poor response to SRL.

00:22:57

SHEEHAN SYNDROME

- Anterior pituitary infarction d/t postpartum hemorrhage
- Chronic presentation > Acute presentation

00:27:03



HORMONAL CHANGES	• GH - 1st to fall ◦ ↓ GH → Hypoglycemia → Emotional lability • ↓ ACTH → Cortisol ↓ → Hypoglycemia • ↓ LH } Secondary amenorrhea/Infertility • ↓ FSH } • ↓ TSH → ↓ T3 & T4 (2° Hypothyroidism) → Weight gain, Constipation, Alopecia • ↓ PRL - Failure of lactation
NEUROIMAGING	• MRI head reveals damage to the pituitary gland → Prolactin ↓
MANAGEMENT	• Hormone replacement ◦ 1st - Dexamethasone → Cortisol ↓ ◦ Combined OCPs: Estrogen (prevent osteoporosis) and progesterone (withdrawal bleeding) ◦ Levothyroxine for ↓ TSH, T4 & T3 ◦ Fludrocortisone → not required for management ◦ In male - Give Testosterone.

SIMMONDS DISEASE

00:34:54

- Non-obstetric cause of damage to the anterior pituitary
 - AVM (Arteriovenous Malformation) rupture
 - Hypertension
 - Sickle cell disease
 - Trauma
- Panhypopituitarism - All 6 hormones from pituitary will be reduced.

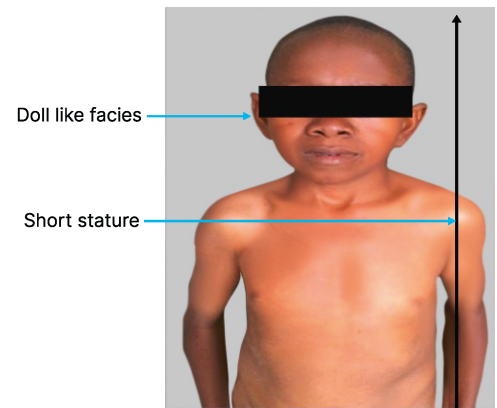
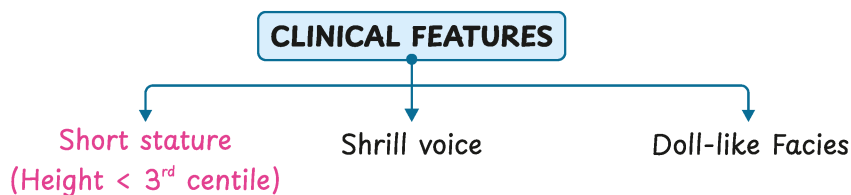
Important Information

- Sipple syndrome is associated with MEN 2A

HYPOPITUITARISM

00:36:43

- Only GH levels reduced



INVESTIGATIONS

- **Insulin challenge test** - no rise in GH level even when sugar level ↓ d/t pituitary defect
 - Risky test d/t chances of hypoglycemia
- **Arginine challenge test** - safer test



o 4G Injection (DNA Recombinant technology) till Puberty

CONFORMATORY TESTS FOR VARIOUS ENDOCRINE DISEASES

00:41:04

ENDOCRINE DISORDER	TEST
Acromegaly	Glucose challenge test
CONN syndrome	Saline infusion test / Salt loading test
Addison disease	ACTH stimulation test/ Cosyntropin test
Diabetes insipidus	Water deprivation test, loss of pituitary bright spot -MRI head
SIADH	Water loading test
Hypopituitarism	Arginine challenge test
Cushing disease	Midnight salivary cortisol levels or Low dose dexamethasone suppression test
Pheochromocytoma	Plasma free metanephrine levels

PROLACTINOMA

00:45:55

- M/c **functioning pituitary tumor** → Produce excess levels of prolactin.
- M/c Pituitary tumor (overall) - Non-functioning adenoma
- Dopamine → inhibits prolactin production → inhibitory effect on LH and FSH

CLINICAL FEATURES	• Anovulation, 2° Amenorrhea/ Infertility • Visual deficits - o Headache
INVESTIGATIONS	• Serum Prolactin levels > 200 microgram/L - suggestive • MRI head - pituitary adenoma
TREATMENT	• Cabergoline (long acting) • Bromocriptine - If pregnancy is desired } ↑ level of dopamine

MCQs

00:51:26

Q. Which prolactin levels are indicative of Prolactinoma?

- > 50µg/L
- >100µg/L
- >150µg/L
- >200µg/L

Ans (d)

Free By - ZeroTrace

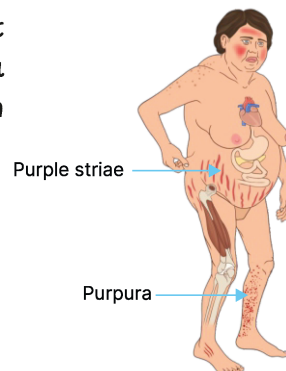


13. ENDOCRINOLOGY IMAGE BASED QUESTIONS

Q. A 45-year-old lady presents with difficulty in climbing stairs, significant weight gain and few cosmetic problems as shown in the image. Her midnight plasma cortisol value is elevated. All of the following are correct about this condition except?

- a. Hyperpigmentation in ectopic ACTH production
- b. Purplish striae
- c. Hyperkalemia
- d. Metabolic alkalosis

Ans: (c)



Inference:

- ↑ midnight plasma cortisol values
- Symptoms are respective to **Cushing syndrome**
- Centripetal obesity
- Stretch marks at the abdomen - purple striae
- ACTH has a partial melanocyte stimulating hormone like action
- ↑ Cortisol → excess stimulation of epithelial Na (ENaC + +) → ↑ reabsorption salt/water → ↑ HTN → urinary loss of K^+ / H^+ → metabolic alkalosis and hypokalemia

Important Information

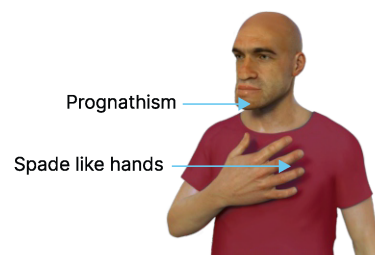
Causes of Hypokalemic alkalosis

- Conn syndrome
- Liddle syndrome
- Cushing syndrome

Q. A 25 year old man with spade like hands and following physical appearance is best diagnosed with?

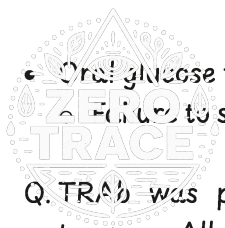
- a. Oral glucose tolerance test
- b. 24 hour urinary GH levels
- c. GHRH stimulation test
- d. Plasma IGF-1 levels

Ans: (a)



Inference

- Spade like hands + prognathism (from image) → **Pituitary adenoma**
- Pituitary adenoma → production of excess GH → acromegaly
- Plasma IGF-1 → screening test for acromegaly
- GHRH stimulation → assessing Hypothalamic pituitary axis

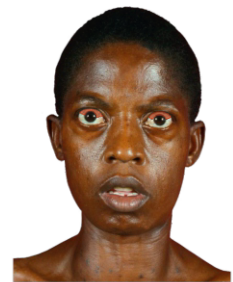


- Oral glucose tolerance test
- Failure to suppress GH production d/t over secretion from pituitary adenoma

Q. TRAb was performed in this patient with weight loss, anxiety and tremors. All of the following will be seen in the condition this patient shown below except?

- Alopecia
- Thyroid storm
- Atrial fibrillation
- Blindness

Ans: (a)

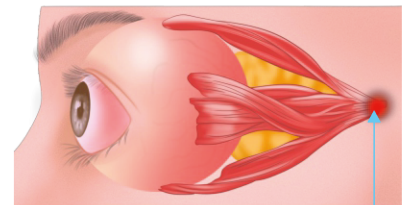


Visible upper part of the sclera

Inference

- TRAb – TSH receptor antibody
- Diagnosis – **Grave's disease**
- $\uparrow T_4 \rightarrow$ sympathomimetic symptoms (weight loss, anxiety, tremors)
- Graves disease can exhibit:
 - **Thyroid storm** (d/t intercurrent illness, MI, stroke, infection, administration of radio iodine when not euthyroid)
 - **Atrial fibrillation** (d/t excess of sympathomimetic stimulation)
 - **Blindness** (d/t retrobulbar neuritis)
- Alopecia is related to myxedema in hypothyroidism

Bulging of Extra ocular muscles leading to proptosis



Retrobulbar neuritis

Q. A patient presents with daily headache, hypertension and muscle cramps. Work up shows low serum potassium. CT abdomen was performed. What is the probable diagnosis?

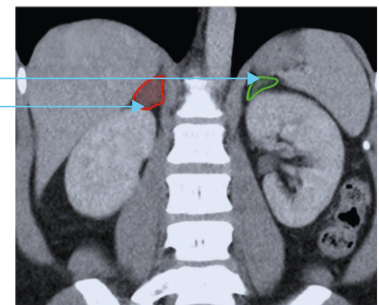
- Pheochromocytoma
- Conn syndrome
- Liddle syndrome
- Secondary hyperaldosteronism

Ans: (b)

Inference

- **Hypertension, muscle cramps $\rightarrow$ aldosterone excess**
- Adrenal mass on the left side
- Pointers against pheochromocytoma
 - **Muscle cramps**
 - **Hypokalemia**
- Liddle syndrome: ENaC gain of function, (-) adrenal mass
- Secondary hyperaldosteronism
 - Pregnancy induced hypertension
 - Barter syndrome
 - Gitelman syndrome
 - Ascites

Normal Adrenal gland
Adrenal adenoma



Q. A 25-year-old presents with weakness, craving for salty foods and postural hypotension. On physical examination, hyperpigmentation over the knuckles is present. Lab investigations show sodium- 130 meq/dl and serum potassium of 6.0 meq/dl, fasting blood sugar of 65 mg%. All are possible causes of the condition shown except.

- a. Histoplasmosis
- b. Amyloidosis
- c. Simmond's disease
- d. Autoimmune polyglandular syndrome type-1

Ans: (c)



Inference

- Weakness, craving for salt, postural hypotension → ↓ aldosterone (salt wasting)
 - Hyperpigmentation over oral mucosa and knuckles → ↑ ACTH
 - Hyponatremia, hyperkalemia, low fasting blood sugar → ↑ cortisol
 - Secondary Addison's disease → pituitary is damaged
 - Histoplasmosis } Can cause damage to the pituitary gland
 - Amyloidosis }
 - Autoimmune polyglandular synd
 - **Simmond's disease:**
 - Non obstetric cause of damage to the pituitary
 - ACTH ↓ → pale skin
- } Primary Addison's disease



ELECTROLYTE IMBALANCES

HYPONATREMIA

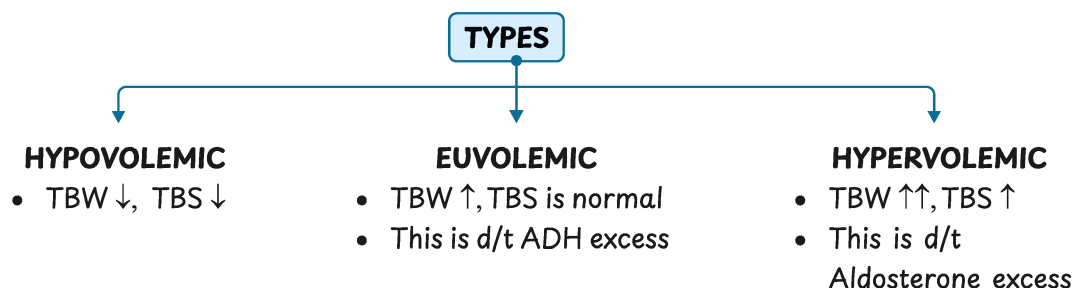
00:00:21

- M/c electrolyte imbalance in hospitalized patients
- Normal serum Sodium: **135-145 mEq/L**
- Cut off for severe Hyponatremia (that can lead to seizures): **<125 mEq/L**
- Plasma osmolality formula
 - Plasma osmolality
 - Equation: $2(\text{Na}^+ + \text{K}^+) + \frac{\text{BUN}}{2.8} + \frac{\text{Glucose}}{18}$
 - Value: **275-295 mOsm**

TYPES OF HYPONATREMIA

HYPOTONIC HYPONATREMIA

- Plasma osmolality <275 mOsm



CAUSES

Hypovolemic Hyponatremia	• Causes ◦ Diarrhea ◦ Vomiting ◦ Fistula ◦ Cerebral Salt Wasting Syndrome → SAH: Excess release of BNP → Natriuresis → Salt ↓ & Water ↓ ◦ Excessive Diuretics • In Diarrhea, vomiting and fistula → Urinary sodium values ↓ • In Cerebral salt wasting syndrome, excessive diuretics → Urinary sodium values ↑
Euvolemic Hyponatremia	SIADH • S - Small Cell Carcinoma of lung, Carcinoid syndrome ◦ SCC: Causes ectopic production of ADH ◦ Incidence of SIADH caused by SCC is higher than incidence of Cushing's syndrome caused by SCC • I - Infection: CNS infections → Reset Osmostat (resetting of threshold of releasing ADH)